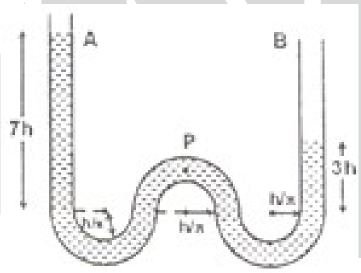
TOPIC : FLUIDS & PROPERTIES OF MATTER

SUBJECT: PHYSICS **DATE: 09-04-2024**

1. A tube of small and uniform cross – sectional area contains water (as shown in the figure) which is initially kept at rest. Now liquid is released. Curved parts are semi-circular with radius $\frac{h}{\pi}$. Pick the correct options for a small sample of fluid of mass dm located at P when level difference reduces to 2h from 4h in the limbs A and B. (Neglect surface tension and viscosity)



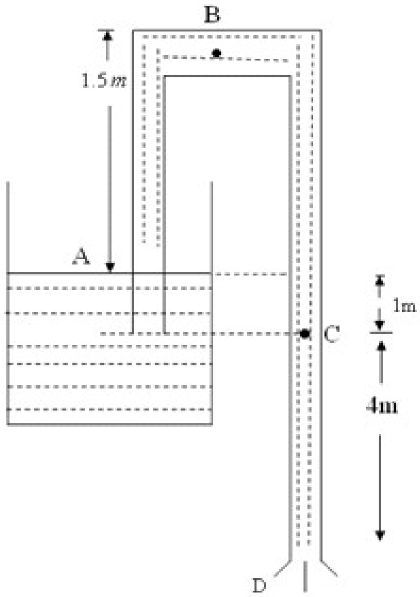
(A) The speed at P is equal to $\sqrt{\frac{8gh}{13}}$

(B) The speed at P is equal to $\sqrt{\frac{6gh}{13}}$

(C) Force on the fluid sample at P is $\frac{2g(dm)}{13}\sqrt{9\pi^2+1}$

(D) Force on the fluid sample at P is $\frac{2g(dm)}{13}$

2. A thin, uniform siphon tube is discharging a liquid of density $900\ kg/m^3$ as shown in figure. [Atmospheric pressure $P_0 = 1.01\times10^5\ N/m^2$]



(A) Speed of liquid through the Siphon is 10 m/s

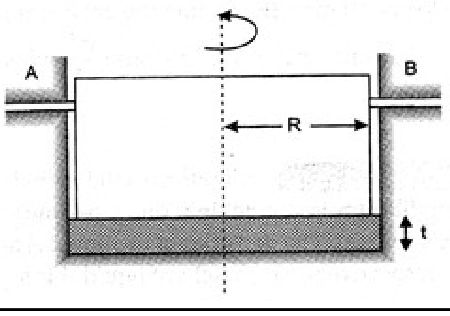
(B) Pressure at point B inside the tube is $4.25\times10^4\ N/m^2$

(C) Pressure at point C inside the tube is $6.5\times10^4\ N/m^2$

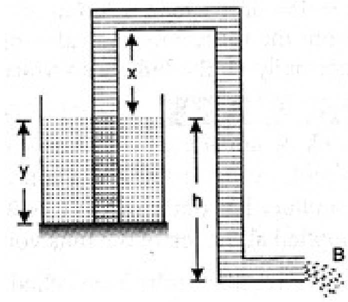
(D) Pressure at point C inside the tube is $8.0\times10^4\ N/m^2$

3. In a certain centrifuge, a dilute water solution of density d_1 with the coefficient of viscosity η of tiny spherical particles of radius r and density $d_2(>d_1)$ is rotated at N rps at a radius R from the axis of rotation. The terminal speed of the particle in the solution is proportional to r^x. Find the value of x

4. A vertical cylinder of radius 'R' is rotating with a constant angular velocity 'ω' along with holders A and B about the axis as shown. The vertical movement of cylinder is restricted by the holders. Oil is only in between the bottom of cylinder and surface as shown. The thickness of the oil layer is 't'. Assume that coefficient of viscosity is 'η' and that cylinder can only rotate if there is no friction elsewhere. Find the power required to overcome the viscous resistance.

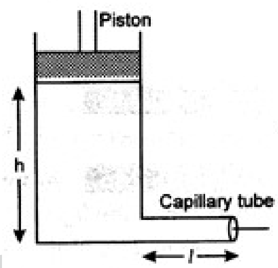


5. A cylindrical tank 0.9 m in radius is filled with water up to a height y=3.0m. A pipe having cross-sectional area 6.3 cm^2 is used to siphon from the tank. End B of the pipe is open to atmosphere and h=5.0 m.($\rho = 1x10^3 kg/m^3$)

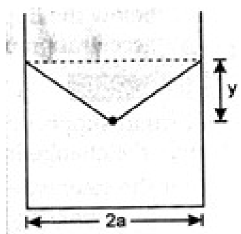


(a) How should the value of *x* be adjusted so that whole of the water can be drained from the tank?
(b) With maximum possible value of *x*, how much time will it take to empty the vessel? (Atmospheric pressure = $1.0x10^5$ Nm^{-2})

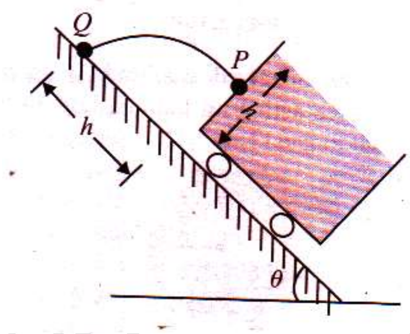
6. An incompressible viscous liquid of density ρ and viscosity η is filled into a cylindrical vessel of cross-section A, up to a height h. The vessel has a horizontal capillary tube of length *l* and bore radius a attached to the bottom of the vessel as shown in the diagram. A piston of mass m is placed on top of the liquid in the tank and the liquid flows slowly and steadily out of the tank under pressure. Find the level of the liquid in the tank as a function of time t. Ignore the surface tension of the liquid.



7. A container of width 2a is filled with a liquid. A thin wire weight of per unit length λ is gently placed over the liquid surface in the middle of the surface as shown in the figure. As a result, the liquid surface is depressed by a distance y(y<<a). Determine the surface tension of the liquid.



8. A cubical tank carrying a non-viscous liquid (water, say) moves down a smooth incline plane freely. When we open the hole made at the mid-point P of the face of the tank, it strikes the point Q of the inclined plane. Speed of the tank at the time of opening the valve is



(A) $\sqrt{gh\cos\theta}\left[2+\frac{\tan\theta}{2}\right]$

(B) $\sqrt{gh\cos\theta}\left(\frac{\tan\theta}{2}\right)$

(C) $\sqrt{gh\cos\theta}\left(1+\tan\theta\right)$

(D) 0

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